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PI Name:	Ianzini, Fiorenza Ph.D.		
Project Title:	Role of high-LET radiation-induced mitotic catastrophe in mutagenesis: implication for carcinogenesis		
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Comments:			
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Flight Program:			
Flight Assignment:	NOTE: Received NCE until 5/31/2011 per PI and C. Guidry/JSC (02/03/2011) NOTE: End date changed to 11/30/2010 per S. Krenek/JSC (7/2010) NOTE: End date change to 9/30/2010 (from 11/30/2009) per JSC info (12/2009)		
Key Personnel Changes/Previous PI:	No changes		
COI Name (Institution):	Mackey, Michael (University of Iowa)		
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Task Description:	This Project aims at testing the hypotheses that iron and proton ions induce genetic imbalances in exposed human cells via processes occurring in cells that undergo radiation-induced mitotic catastrophe (MC) (reviewed in Ianzini and Mackey 2007) and that this induced instability correlates with increased cancer risk. In previous Task Reports we have discussed data relative to the effects of iron ions and proton beams irradiation on human fibroblast cells. During this past year we have begun irradiation experiments with epithelial cells, the NHBE human bronchial/tracheal cells, exposed to different doses of 1 GeV iron ions.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	The studies conducted in this project fell in the category of risk #28 – Carcinogenesis - of the Radiation Health Bioastronautics Roadmap Table. In particular, we are addressing the following questions: 28b. How can tissue specific probabilities for increased carcinogenesis risk from space radiation be best evaluated? What molecular, genetic, epigenetic, abscopal (effect that irradiation of a tissue has on remote non-irradiated tissue), and other factors contribute to the tissue specificity of carcinogenic risk? To answer this question we are testing the hypothesis that high-LET radiation leads to an enhanced incidence of MC (definition and description of MC is reported in the “Task Description” section). The idea is to define MC susceptibility of human cells exposed to densely ionizing radiation to be used as a “biomarker” for delayed effects of exposure to high-LET radiations 28i. What are the probabilities that space radiation will produce DNA damage at specific sites, including clustered DNA damage? What is the likelihood that DNA damage will increase the risk of carcinogenesis? To answer this question we are testing the hypothesis that DNA damage is persistent in survivors of cells that have undergone radiation-induced-MC and that this genetic imbalance increases cancer risk. 28j. What mechanisms modulate radiation damage at the molecular level (e.g., repair, errors in repair, signal transduction, gene amplification, bystander effects, tissue microenvironment, etc.) that significantly impact the risk of cancers, modulate the expression of radiation damage and decrease the radiation risk, and how can the understanding of mechanisms be used to predict carcinogenic risks from space radiation? To answer this question we are following the irradiated cell populations for long time post-irradiation by means of live cell imaging. Our previous observation using the Large Scale Digital Cell Analysis System (LSDCAS) – an in-house built live cell imaging system extremely versatile – has shown that a small proportion of irradiated cells undergoing MC escape death. We are following these survivors to determine their viability. If these cells form viable colonies and are persistent for long time the risk of cancer development under these conditions can be deemed increased.		

Task Progress:	SUPPLEMENTAL INFORMATION for FINAL REPORT PROGRESS, dated August 2011 One of the most important findings of our studies is that we were able to demonstrate that all the cell lines we have analyzed (murine or human, embryonic, fetal or adult) undergo radiation-induced MC and that they are able to survive for weeks post-irradiation. These results, coupled with findings on cancer cells (Ianzini et al. 2009) that demonstrated that irradiated cells undergo MC, become polyploid, and are able to escape death by activating processes of depolyploidization that utilize uncharacteristic pathways of cell divisions (namely activation of meiotic genes during multipolar divisions), prompted us to hypothesize that similar processes might occur in normal cells exposed to space radiation. The potential occurrence of these processes, associated with the production of persistent DNA damage, is of great importance as it would imply that normal cells that receive a dose of space radiation and undergo MC can escape death by re-organizing their genomic composition. If these cells are also able to proliferate (as we have demonstrated being the case for a small percentage of cancer cells that undergo depolyploidization – Ianzini et al. 2009), their progeny would likely have a disarrayed DNA that could be prone to further mutations and ultimately render this cell progeny more susceptible to carcinogenesis. Believing that this reasoning would broaden our understanding of specific pathways of survival in irradiated normal cells we devoted our last year and a half of funding towards demonstrating that normal cells not only undergo radiation-induced MC and escape death, but also become polyploid and express meiosis-specific genes possibly involved in the depolyploidization process detected through live cell imaging that may play a role in long-term cell survival (Kosmacek et al. 2011). We have found that both meiosis-specific genes and specific markers of cancer stem cells are expressed in irradiated polyploid normal cells of different origin at times matching the multipolar divisions measured in live cell imaging experiments. These data are instrumental to our understanding of the long-term detrimental effects of radiation as it pertains to cancer risk. During this funding period, we also used some of the irradiation time that was allotted to us to acquire preliminary data for a second project within NASA interests [effects of space radiation on central nervous system (CNS)]. This project is a collaborative effort between the University of Iowa and the Loma Linda University Medical Center and aims at testing the hypothesis that neural stem cells (NSC) exposed to different energies of proton and iron ions are functionally altered, effectively resulting in decrements of the NSC ability to migrate, proliferate, integrate and reduce injury within the brain of an animal model of trauma. Within this project we have gained preliminary data on DNA damage and motility characteristics of NSC. Annual Report for Year 2010 - compiled in february 2011 - Epithelial cells exposed to iron ions undergo radiation-induced mitotic catastrophe. Live cell imaging analysis of NHBE cells demonstrates that iron ion irradiated NHBE cells are often unable to complete cell division, an event that promotes cell fusion, with the production of large binucleated cells; cell fusion is a hallmark of MC. Moreover, iron ions induce changes in the NHBE phenotype evidenced by the augmented size of the irradiated cells compared to the untreated control and by the slowed growth. Thus, NHBE cells, while seem to be resistant to multiple nuclear fragmentation, undergo MC events by becoming large binucleated cells, and presenting changes in morphology, cell size and growth/division properties. Epithelial cells that have undergone iron ion-induced MC are positive for gamma-H2AX. DNA damage studies, conducted on the NHBE cells show that the expression of gammaH2AX protein is time- and dose-dependent. In irradiated cells gamma -H2AX is promptly recruited at sites of DNA dsb within the first 73 h to repair the radiation initial damage. Gamma-H2AX does not disappear after having accomplished its first response to radiation damage, but remains expressed at various degrees for the duration of the entire experiment (up to day 15 post-irradiation). Moreover, at the later times post-irradiation some of the cells have become smaller and only the large multinucleated cells express gamma -H2AX. Thus, it appears that foci of H2AX recruitment are persistent in the irradiated cells that present MC morphology. Control cells are small and negative for gamma -H2AX expression at any time during the course of the experiment. Epithelial cells express meiosis-specific genes at later times post-iron ion radiation exposure. Because the results discussed above demonstrate the presence of a heterogeneous irradiated cell population where small (mononucleated) cells and large (multinucleated) cells are found at later times post-irradiation and where only the large multinucleated cells express phosphorylated H2AX, we hypothesized that the small mononucleated cells might be the result of depolyploidization events occurring in the large multinucleated cells. To test this hypothesis we searched for markers of depolyploidization, that is genomic reduction. In nature genomic reduction occurs via reduction division, and reduction division (or meiosis) is under the strict regulation of sets of specific genes. Thus we began testing the expression of gene products that are only expressed during meiosis and that are in particular involved in chromatid pairing and homologous recombination. To this end we investigated the expression of SYCP3 (synaptonemal complex - protein 3), a meiosis-specific proteinaceous structure that develops during pairing of homologous chromosomes and mediates chromosome pairing, synapsis, and crossover. In irradiated NHBE cells SYCP3 expression is dose- and time-dependent, with fewer cells expressing SYCP3 at lower doses, and with more cells staining positive for SYCP3 starting after day 3 post-irradiation. To note is the fact that the multinucleated cells are those expressing SYCP3, not the smaller mononucleated cells. It is remarkable that at the same time when NHBE cells express gamma -H2AX, they also express SYCP3. These data suggest that SYCP3, by facilitating chromosome interchanges, and H2AX, by repairing DNA open ended backbone may aid the cells to become more genetically stable and more suitable for overall cell survival. Epithelial cells express self-renewal genes at later times post-iron ion radiation exposure. Along with SYCP3, irradiated NHBE cells also express the cancer stem cell marker nestin. Nestin is a Class VI intermediate filament protein that regulates the assembly and disassembly of intermediate filaments in the developing central nervous system and participates in cell remodeling . Expression of SYCP3 and nestin in irradiated NHBE cells is dose- and time-dependent, with onset at day 7 post-irradiation for the cells irradiated with doses of 0.5 and 1.0 Gy of iron ions and at day 5 post-irradiation for the cells irradiated with dose of 2 Gy. To note is the fact that SYCP3 and nestin are again expressed by the large multinucleated cells rather than by the smaller cells in the population. These findings suggest that the activation of meiosis-specific genes and of self-renewal proteins is elicited at times of depolyploidization, possibly to govern a more stable distribution of genetic material within the daughter cells that would allow for the cells' successful proliferation. Overall considerations. These findings support the hypothesis that normal cells that have undergone radiation-induced MC, but escaped death, retain proliferative capabilities facilitated by the activation of unusual pathways of cell division that utilize meiosis-specific and self-renewal gene products that endow the irradiated cells with genetic and/or phenotypic characteristics of cancer stem cells. Thus, we believe it possible that aberrant expression of stem cell genes may occur via the activation of silenced gametogenic programs and that this switch may play a role in carcinogenesis. By conducting experiments of anchorage-independent cell growth, we are at the moment testing the hypothesis that the irradiated NHBE cells that escape radiation-induced MC death acquire tumorigenic potential.
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